



Programming Fundamentals

Conditional Statements

Programming Fundamentals Team

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Outline

1. Boolean Conditions
2. Logical Operators
3. Common Syntax Errors



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1. Boolean Conditions

Recap

- A Boolean condition is an expression that evaluates to either **True** or **False** e.g.

```
x < 50  
number > 0
```



1. Boolean Conditions

Recap

- A Boolean condition is an expression that evaluates to either **True** or **False** e.g.

```
x < 50  
number > 0
```

- A conditional statement evaluates a Boolean condition and its result will determine which section of the code is executed.



1.1 Relational Operators

Operator	Syntax	Returns true if...
>	x > y	x is greater than y
>=	x >= y	x is greater than or equal to y
<	x < y	x is less than y
<=	x <= y	x is less than or equal to y
==	x == y	x and y are equal
!=	x != y	x and y are not equal



1.1 Relational Operators

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!=	x != y	x and y are not equal

= is an assignment operator and doesn't test for equality.
== is used to test for equality.



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2. Logical Operators

- Logical operators operate on Boolean values and produce a new Boolean value as a result.



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- Logical operators operate on Boolean values and produce a new Boolean value as a result.
- The three logical operators in Python are:

Operator	Meaning	Example
and	True if both conditions are true	<code>age >= 18 and score > 50</code>
or	True if at least one condition is true	<code>day == "Sat" or day == "Sun"</code>
not	Reverses the Boolean value	<code>not is_raining</code>



2.1 and operator

a and b

a	b	a and b
True	True	True
False	True	False
True	False	False
False	False	False



2.1 and operator

a and b

a	b	a and b
True	True	True
False	True	False
True	False	False
False	False	False

- Evaluates to **true** only if **both** a and b are true.
- It is false in all other cases.

```
age = 25
income = 35000

if age >= 18 and income >= 30000:
    print("Eligible for loan")
else:
    print("Not eligible")
```



2.2 or operator

a or b

a	b	a or b
False	False	False
False	True	True
True	False	True
True	True	True



2.2 or operator

a or b

a	b	a or b
False	False	False
False	True	True
True	False	True
True	True	True

- Evaluates to **true** if either a, b, or both are true.
- It is false only when **both** are false.

```
day = "Saturday"

if day == "Saturday" or day == "Sunday":
    print("It is the weekend!")
else:
    print("It is a weekday.")
```



2.3 not operator

not a

a	not a
False	True
True	False



2.3 not operator

not a

a	not a
False	True
True	False

- Reverses (negates) the Boolean value.
- True becomes False, and False becomes True.

```
is_raining = False

if not is_raining:
    print("Leave your umbrella at home.")
else:
    print("Don't forget your umbrella!")
```




2.4 Logical Operators — Quiz

Given these variables:

```
a = 5      b = 10      c = 7
```

What is the result of each Boolean expression?

Q	Expression
Q1	<code>(a > b) and (a < c)</code>
Q2	<code>(a < b) or (c < a)</code>
Q3	<code>not(b < a) and (c > b)</code>



2.4 Logical Operators – Quiz

Answers:

```
a = 5    b = 10    c = 7
```

Q	Evaluation	Result
Q1	(5 > 10) is False . (5 < 7) is True . So False and True evaluates to False .	False
Q2	(5 < 10) is True . With or, one true operand is enough — overall answer is True .	True
Q3	not(10 < 5) = not(False) = True . (7 > 10) is False . So True and False evaluates to False .	False



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3.1 Splitting a long Boolean condition incorrectly

Problem — splitting a long condition across lines:

```
age = 25
income = 35000

if age >= 18 and
    income >= 30000:
    print("Eligible")
```

SyntaxError: invalid syntax



3.1 Splitting a long Boolean condition incorrectly

Problem — splitting a long condition across lines:

```
age = 25
income = 35000

if age >= 18 and
    income >= 30000:
    print("Eligible")
```

SyntaxError: invalid syntax

Fix — wrap the full condition in parentheses:

```
age = 25
income = 35000

if (age >= 18 and
    income >= 30000):
    print("Eligible")
```

Parentheses allow the condition to span multiple lines.





Thanks for Watching - Any questions?

